

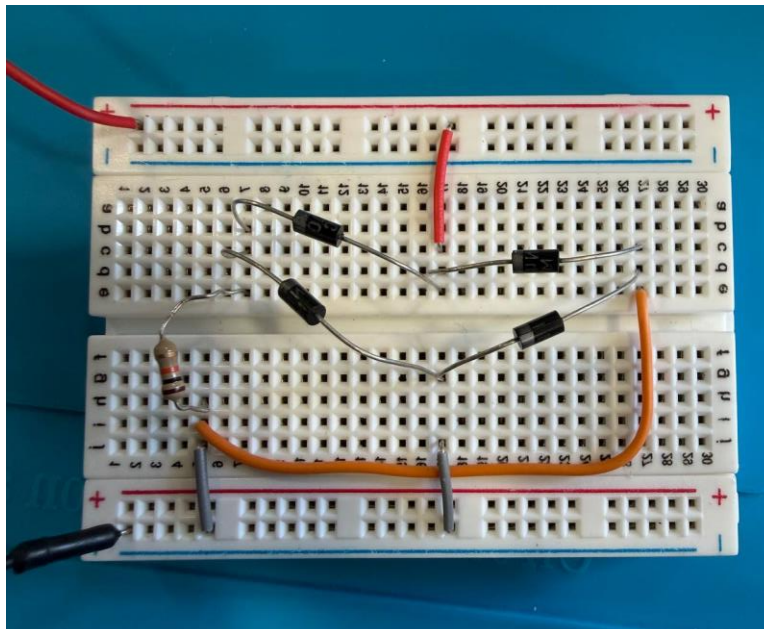
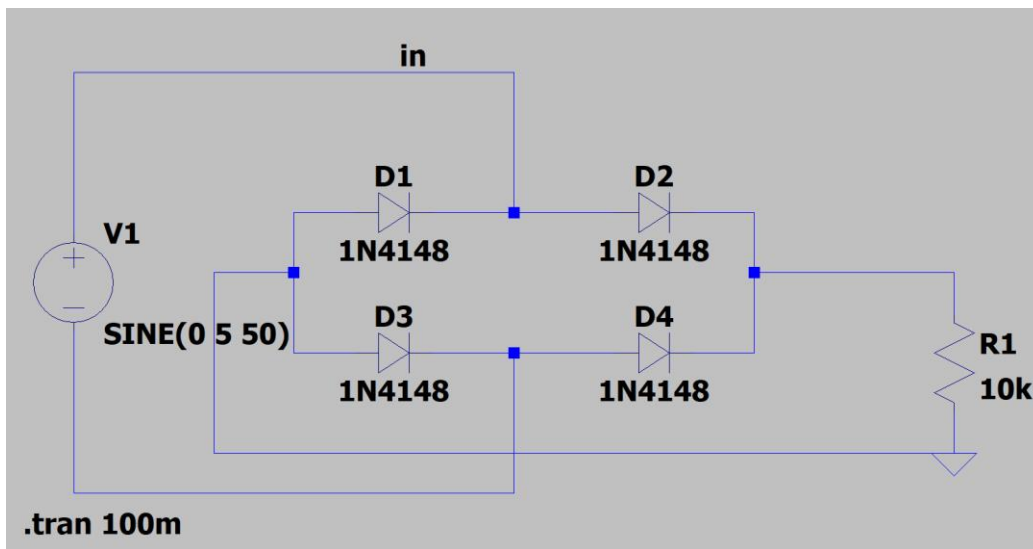
Diode Rectifier

ENGS152 Circuits Lab

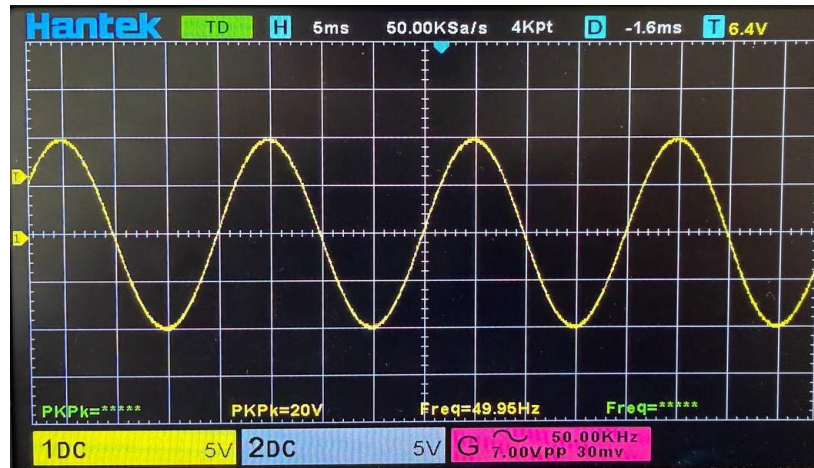
22 March 2026

Step 1: Full-wave rectifier

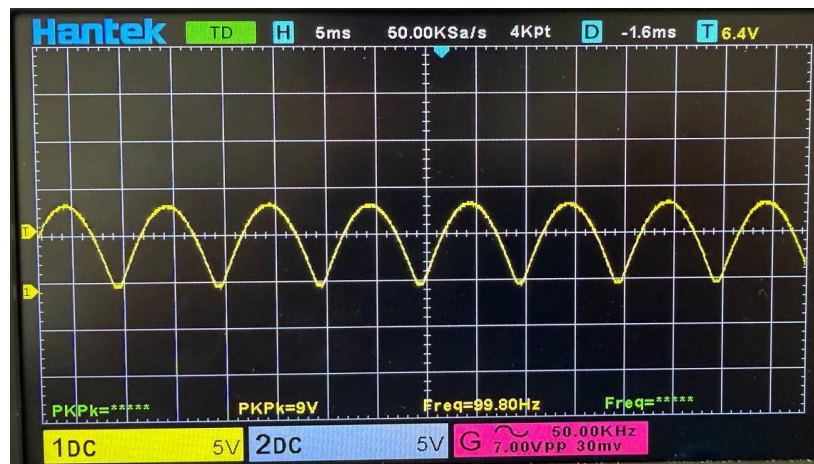
First, we are asked to build a full-wave rectifier circuit using 1N4001 diodes and a 10k Ω load resistor. You can see the schematic circuit and real-life assembly on the breadboard below.



For this lab an external signal generator has been used to create a sine wave with an amplitude of 20 V. The frequency of the generated sine should be 50 Hz. Below you can see the input voltage before entering the rectifier.



After passing through the full-wave rectifier this is how the voltage plot looks like. As you can see the circuit flips the negative phase of the sine wave and only outputs the positive part of the voltage. The frequency doubles and the peak voltage is lower than the input.



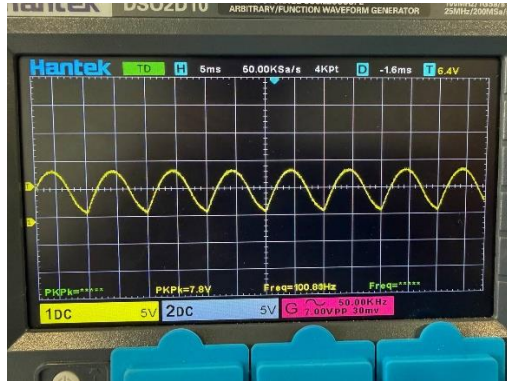
As you can see the Peak-to-Peak voltage is 9V. Ideally without any resistance in the diodes the Peak-to-Peak voltage would be half the input voltage 10V, however diodes have their inner resistances. The voltage drop for two forwards diode is:

$$V_{drop} = V_{In} - 9V = 1V$$

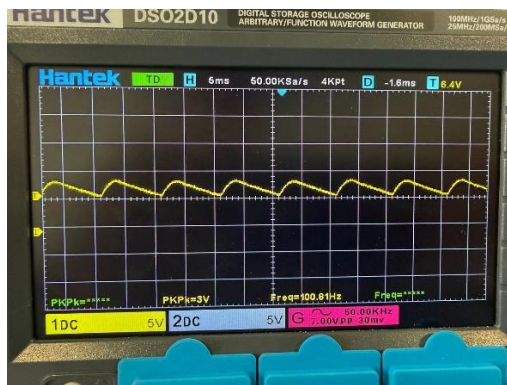
$$\text{Voltage drops through each diode: } V_{diode} = \frac{1}{2} = 0.5V$$

Next up, we are asked to connect $0.22\mu\text{F}$, $2.2\mu\text{F}$, $22\mu\text{F}$ capacitors in parallel to the load resistor and observe the behavior of the output voltage.

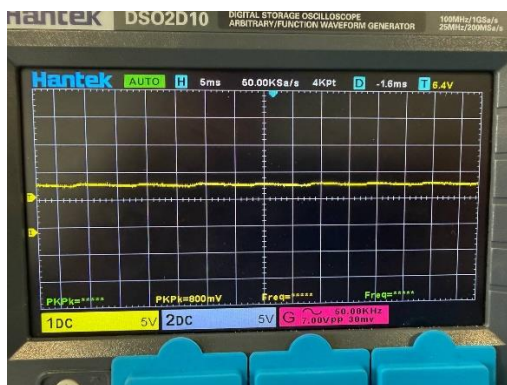
$0.22\mu\text{F}$:



$2.2\mu\text{F}$:



$22\mu\text{F}$:

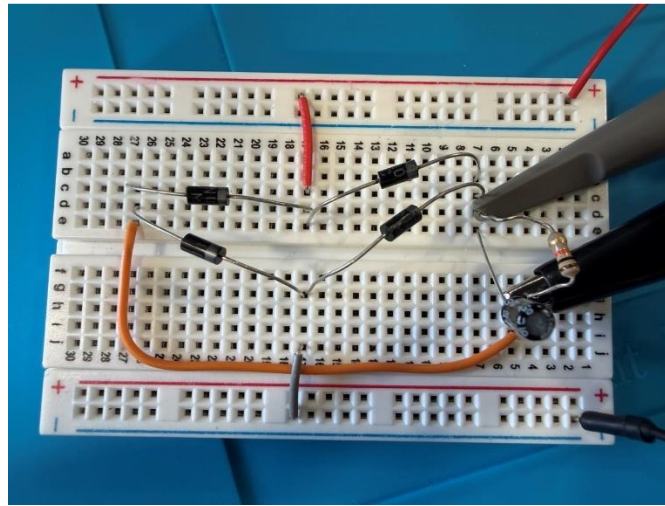


As capacitance values increase, the capacitor store more charge and it takes longer time to discharge, reducing the pikes in the voltage and making the voltage a smoother DC output, and the voltage at the output significantly drops because the capacitor discharges through the load when the input voltage is decreasing.

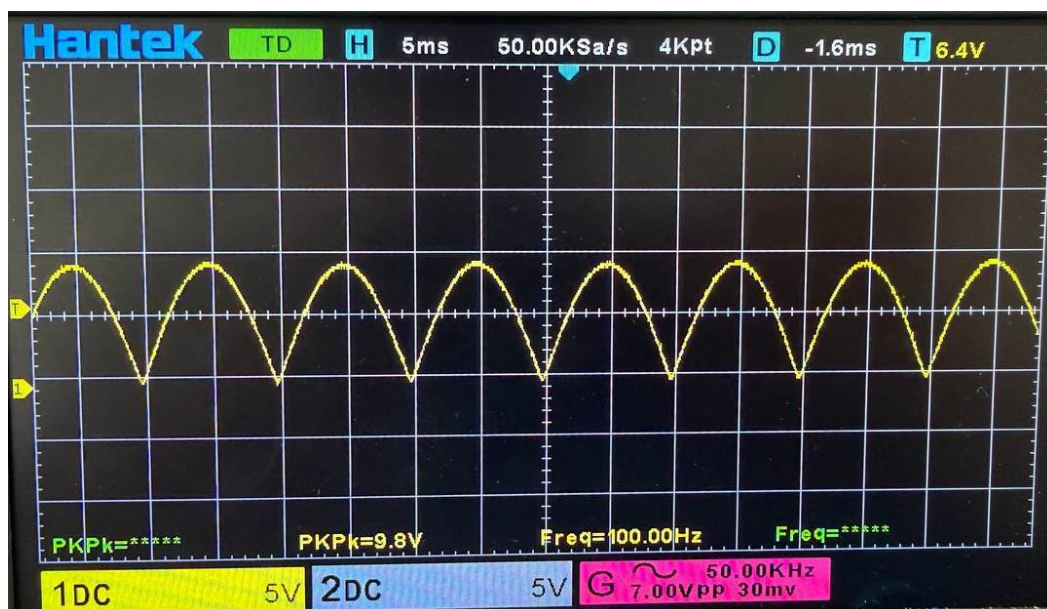
Step 2: Full-wave rectifier using Schottky diodes

In this part of the lab, we are asked to substitute the 1N4001 diodes with 1N5817 Schottky diodes. The difference between these two diodes is that the Schottky diodes are more efficient, hence they lose less voltage.

The schematic circuit and real-life assembly on the breadboard look pretty much like the ones in the previous step of the lab. The only difference is that different diodes have been used and all other connections are the same.



Below you can see the rectified output through the full-wave rectified circuit with Schottky diodes.



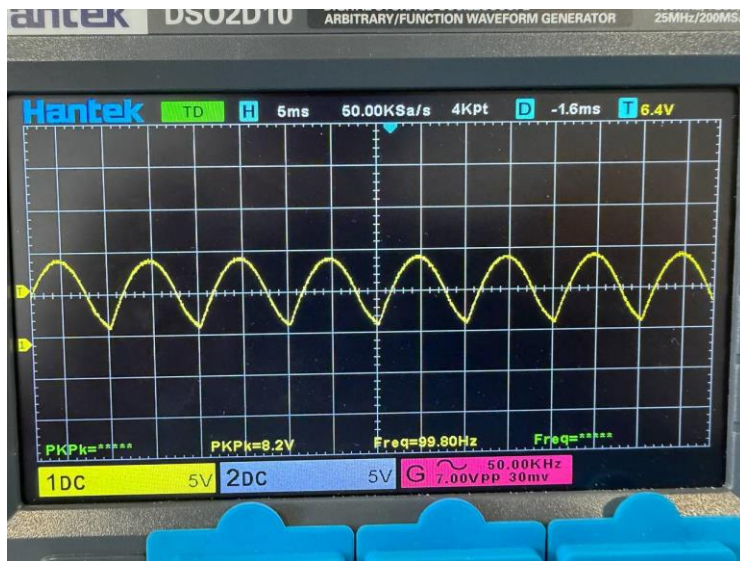
As you can see the voltage drop is significantly less than the one observed in the previous step of the lab.

$$V_{\text{drop through each diode}} = \frac{10 - 9.8}{2} = 0.1V$$

This proves the claim that Schottky diodes are more efficient than the regular ones.

Just like in the previous step of the experiment we are asked to connect $0.22\mu\text{F}$, $2.2\mu\text{F}$, $22\mu\text{F}$ capacitors in parallel to the load resistor and observe the behavior of the output voltage.

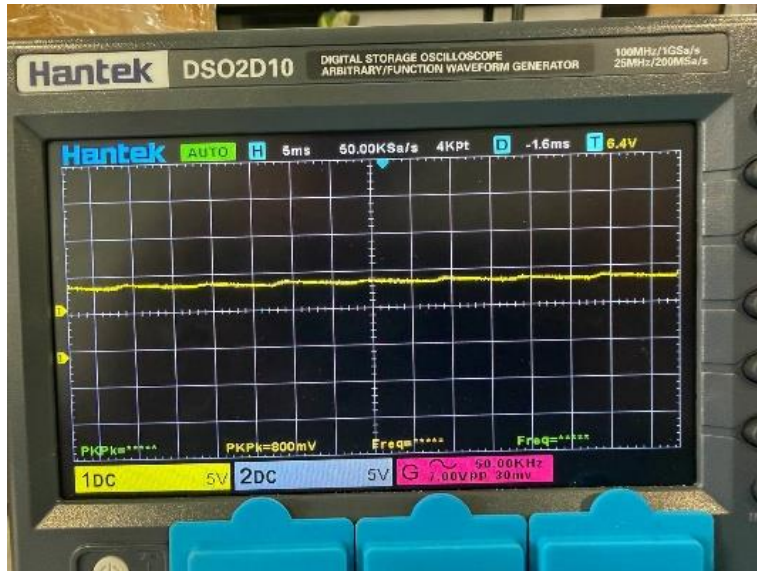
$0.22\mu\text{F}$:



$2.2\mu\text{F}$:



22 μ F:



As we can see, the same behavior is observed, as the capacitance increases the output voltage becomes smoother and smoother making it more DC like current. The only difference is that the voltage drop is significantly less than the ones observed in the regular silicone diode circuit. However, for the large capacitance 22 μ F, we can see that both circuits have the same output of 800mV. This is because the capacitor needs more current to charge to high voltage, but the circuit did not have enough time to charge the capacitor. As a result, the capacitor did not charge properly and stayed at low voltage. Hence, both regular diode and the Schottky diode gave almost the same output.